

SHANMUKHA KRISHNA CHAITANYA MUNAGALA

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PROFESSIONAL SUMMARY

MSc Advanced Machine Learning student at the University of Bath with prior industry experience as a Software Engineer (Machine Learning), developing and deploying machine learning solutions on large-scale real-world data. Strong foundation in Python, Scikit-learn, PyTorch, TensorFlow, feature engineering, model development, model evaluation, anomaly detection, and Natural Language Processing (NLP). Published machine learning researcher with hands-on experience across financial data, smart grid systems, battery analytics, and applied classification problems, seeking opportunities in Machine Learning, Data Science, and AI-focused roles.

Research Interests: Applied Machine Learning, Artificial Intelligence, Anomaly Detection, Natural Language Processing (NLP), Deep Learning, Generative AI, Scalable ML Pipelines

EDUCATION

2026 – 2028 (Expected)	Master of Science (MSc) in Advanced Machine Learning (Including Professional Placement Year) University of Bath, Bath, England, United Kingdom
2022 – 2026	Bachelor of Technology (B.Tech) in Computer Science and Engineering Specialization: Artificial Intelligence and Machine Learning SRM University AP, Amaravati, Andhra Pradesh, India GPA: 7.89 / 10

SKILLS

Programming:	Python, C
Machine Learning:	Classification, Regression, Clustering, Anomaly Detection, Ensemble Learning, Feature Engineering, Model Selection, Cross-Validation, Model Evaluation
Deep Learning & NLP:	Neural Networks, Deep Learning, Natural Language Processing (NLP)
Generative AI:	Large Language Models (LLMs), GPT-based Applications, Prompt Engineering, Structured Prompting
Data Science & Libraries:	Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, Matplotlib, Exploratory Data Analysis (EDA)
Tools:	Git, GitHub, Jupyter Notebook, Google Colab

WORK EXPERIENCE

Software Engineer (Machine Learning) Spizen Technologies (CLR3 Ventures), Bengaluru, India	December 2025 – April 2026
- Developed and deployed machine learning models for similarity pattern matching and price classification on large-scale blockchain transactional data in real-time production ML pipelines. - Designed end-to-end ML workflows including data engineering, data preprocessing, feature engineering, model training, and performance evaluation. - Conducted applied machine learning research for pattern discovery and anomaly detection in dynamic financial data systems.	

Machine Learning Research Intern SRM University AP, Amaravati, India	June 2025 – July 2025
- Developed a machine learning pipeline for Smart Grid stability classification using multi-feature operational parameters and evaluated multiple models. [Certificate] - Implemented data preprocessing and class imbalance correction techniques (SMOTE-Tomek) to improve model performance.	

Front-End Developer Indian Institute of Technology Roorkee, India	June 2024 – July 2024
- Developed responsive front-end components for an educational virtual lab using HTML, CSS, and JavaScript [Certificate]. - Collaborated with a multidisciplinary team to integrate theoretical modules, computational tools, and assessment features into an interactive learning platform.	

PUBLICATIONS

- **Shanmukha Krishna Chaitanya Munagala**, et al., "[Thermal Stress based Degradation Classification of Li-ion Batteries Using Machine Learning](#)", *6th IEEE International Conference on Sustainable Energy and Future Electric Transportation (SeFet 2026)*, 08-11 Jul, 2026, VNIT Nagpur, Maharashtra, India.
- Sunkara Ekeswara Reddy, **Shanmukha Krishna Chaitanya Munagala**, "[Detection of Single-Line-to-Ground Faults in Smart Grid using Machine Learning](#)", *Fifth International Conference on Power, Control and Computing Technologies (ICPC2T-2026)*, 11-13 Mar, 2026, NIT Raipur, Chhattisgarh, India.
- Bhagya Satya Sri Vutukuri, **Shanmukha Krishna Chaitanya Munagala**, et al., "[A Machine Learning Approach for Efficiency-Based Fault Detection in EV batteries](#)", *2025 IEEE 4th International Conference on Smart Technologies for Power, Energy, and Control (STPEC 2025)*, 10-13 Dec, 2025, NIT Goa, Goa, India.
- **Shanmukha Krishna Chaitanya Munagala**, et al., "[Performance Analysis of Fire and Smoke Detection System Employing Machine Learning Techniques](#)", *International Conference on Computational Robotics, Testing and Engineering Evaluation (ICCRTEE 2025)*, 28-30 May, 2025, Kalasalingam Academy of Research and Education, Tamil Nadu, India.

PROJECTS

Machine Learning-Based Classification for Smart Grid Stability Using Operational Features Dataset

- Developed a machine learning pipeline for Smart Grid stability classification using a 60,000-instance operational dataset, incorporating preprocessing steps such as feature refinement, stratified splitting, and class imbalance handling using SMOTE-Tomek.
- Trained and evaluated multiple supervised models (Logistic Regression, KNN, Naive Bayes, Random Forest, Gradient Boosting), achieving up to 92.11% accuracy and 0.98 AUC-ROC, with ensemble methods demonstrating superior performance in capturing nonlinear system dynamics.

Keyword Extraction for Content Optimization

- Designed an end-to-end NLP pipeline for keyword extraction using Python, Pandas, NLTK, and Scikit-learn, incorporating data collection and preprocessing to remove over 90% of structural noise.
- Implemented and compared TF-IDF, TextRank, and YAKE algorithms to analyze performance trade-offs and derive data-driven insights for content optimization.

Waste Image Classification

- Developed a multi-class image classification system for waste categorization using machine learning techniques.
- Built a feature extraction and dimensionality reduction pipeline using ResNet50 and Autoencoder, and applied Genetic Algorithm with XGBoost for feature selection and classification, benchmarking performance against SVM and ensemble models.

Epilepsy Diagnosis through Machine Learning

- Developed a machine learning pipeline for epileptic seizure classification using a large-scale EEG dataset.
- Engineered a preprocessing workflow to extract statistical features, reducing data from 3 million to 30,000 samples while preserving critical information.
- Trained and evaluated multiple models, achieving a peak accuracy of 95.03% using XGBoost after addressing class imbalance with SMOTE.

CERTIFICATIONS

DeepLearning.AI: [Neural Networks and Deep Learning](#) in Coursera by DeepLearning.AI
Coursera: [Machine Learning Specialization](#) provided by **Stanford University and DeepLearning.AI**

WORKSHOPS & SEMINARS ATTENDED

Govt of Andhra Pradesh:	Participation in Amaravati Quantum Valley Workshop (Jun 2025)
SRM University AP:	Participation in 9th Research Day (Apr 2025)
IIT Roorkee:	Participation in Workshop on Generative AI By Intel (Mar 2024)
IIT Roorkee:	Participation in Workshop on Basics Of Stock Market By SEBI (Mar 2024)
IIT Madras:	Attended Product Management Workshop (Jan 2024)
IIT Madras:	Attended UI/UX with Figma Workshop (Jan 2024)
IIT Bhubaneswar:	Attended Full Stack Web Development Workshop (Mar 2023)
SRM University AP:	Participation in the workshop on Hands-on BLOCKCHAIN (Mar 2023)